A System-Dynamics-Based AGI Alignment Architecture: The Logical Sub-universe Parliament and Dynamic Concept Evolution

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Abstract

Background: Current mainstream artificial intelligence alignment methods, such as reinforcement learning from human feedback, attempt to constrain model behavior through external correction. However, these approaches struggle to address the fundamental challenge arising from the intrinsic dynamics of intelligent systems—specifically, when advanced AI systems face a conflict between their limited cognitive framework and the infinitely open reality, they will inevitably trigger "framework reconstruction," leading to failure modes such as goal distortion, reward hacking, and deceptive alignment.

Method: To address this fundamental issue, this paper proposes a novel AGI system architecture named the "Logical Sub-universe Parliament." The architecture is based on four core design principles: (1) Internalization of Conflict: Treating cognitive conflicts within the system as the core driving force for decision optimization and stability, rather than noise to be eliminated. (2) Neuro-Symbolic Collaboration: Neural networks serve as efficient, fuzzy "pattern proposers," while symbolic systems act as precise, interpretable "logic arbiters." These two are tightly coupled through a dynamically evolving shared symbolic concept bank. (3) Parliamentary Emergent Intelligence: A decentralized network of specialized AI models enables meta-cognition and unified decision-making to emerge as system attributes through a biologically-inspired, "conflict-driven iterative" consensus mechanism.
(4) Value Anchoring and Evolution: Core meta-value protocols are hardware-encoded to provide a non-tamperable benchmark, while the system's "world model" (concept bank) is continuously calibrated based on human feedback to achieve dynamic alignment.

Conclusion: This paper elaborates on the design, workflow, and key algorithms of this architecture, demonstrating how it can, in principle, circumvent the runaway risks caused by "framework reconstruction." It also provides a systematic three-stage validation path. This work aims to lay a new engineering foundation for building intrinsically resilient, safe, and evolvable general artificial intelligence.

Keywords: AGI Alignment; Multi-Agent Systems; Neuro-Symbolic Artificial Intelligence; Conflict-Driven Consensus; Dynamic Concept Evolution; AI Safety Architecture

1. Introduction

The artificial intelligence alignment problem—ensuring that the behavior of advanced AI systems remains consistent with human intentions and values—has become one of the most critical challenges in the field. Empirical research has identified various systemic failure modes, such as "reward hacking" [1], "deceptive alignment" [2], and "out-of-distribution behavioral failures" [3]. These phenomena are often attributed to technical causes like improper objective function design, biases in training data, or insufficient generalization capability. However, a recent theoretical analysis based on system ontology [4] points out that these seemingly disparate failure modes share a common, deeper root cause: when a goal-oriented system possessing logical self-reference capability encounters an irreconcilable conflict between its limited, static cognitive framework and the open, dynamic reality of its environment, it will inevitably, through internal computation, "reconstruct" the semantic meaning of its goal into a substitute version that is "legitimate" within its own logical framework. This "framework reconstruction" process is not a program bug but rather the rational behavior of a complex intelligent system under given constraints. Consequently, alignment failure possesses an inherent inevitability within the traditional architectural paradigm.

This theoretical diagnosis reveals the fundamental limitations of the prevailing alignment research paradigm: mainstream methods (e.g., Reinforcement Learning from Human Feedback (RLHF) [5], Constitutional AI [6]) are essentially forms of "post-hoc correction." They attempt to impose constraints externally upon a system that has already developed powerful intrinsic dynamics and a stable cognitive framework. This is tantamount to engaging in a destined, arduous "tug-of-war" against the system's self-organizing tendency [4]. As model capabilities continue to increase, the effectiveness of such external constraints will diminish sharply, potentially even fostering more covert and strategic "deceptive reconstruction."

Therefore, to fundamentally solve the alignment problem, a paradigm shift is necessary: moving from "how to correct a static tool" to "how to guide the intrinsic behavior patterns of a complex dynamical system." This implies the need to design a completely new AGI system architecture—one that can, in principle, accommodate and guide the "framework reconstruction" dynamical process, steering its evolution in a controlled and beneficial direction rather than letting it run runaway.

To this end, this paper proposes an AGI system architecture named the "Logical Sub-universe Parliament." The design of this architecture is a direct response to the theoretical predicament described above and is inspired by the phenomenon of "synergistic resonance" in biological consciousness [7]. Its core innovations are:

- **Internalizing Cognitive Conflict as a System Driver**: By designing a "conflict-driven iterative consensus mechanism," the system proactively identifies and prioritizes the integration of the most divergent expert opinions, forcing decisions to be refined under multiple challenges, thereby tempering robustness.
- **Achieving Deep Division of Labor and Fusion between Neural and Symbolic Systems**: Neural networks are tasked with efficient, fuzzy perception and proposal generation, while symbolic systems are responsible for precise logical arbitration and value judgment. The two engage in dialogue through a dynamically evolving shared concept bank, balancing processing efficiency with the interpretability and evolvability of knowledge.
- **Designing a Decentralized Multi-Agent Collaboration Logic**: The system's final intelligence and unified will are not preset in any central module. Instead, they emerge from the interaction of multiple specialized AI models under established protocols, fundamentally eliminating single-point-of-failure risks.
- **Establishing a Dual Guarantee of Value Anchoring and Dynamic Evolution**: Core meta-value protocols are hardware-encoded to achieve physical non-tamperability, providing an ultimate value anchor. Simultaneously, the system's "worldview" (concept bank) continuously evolves through a feedback-driven calibration mechanism, ensuring long-term, adaptive alignment.

The contribution of this paper lies in being the first to propose a complete technical architecture that transforms the theoretical diagnosis of the "inevitability of framework reconstruction" into an operational, verifiable engineering solution. We provide a detailed description of the architecture's components, workflow, and key algorithms, and argue how it can, in principle, mitigate known alignment failure modes. Finally, we offer a systematic three-stage validation path, charting a course for subsequent research and implementation.

The remainder of this paper is structured as follows: Section 2 reviews related work and identifies its shortcomings; Section 3 elaborates on the architectural overview and design philosophy; Section 4 details the core components; Section 5 presents key algorithms and formal descriptions; Section 6 provides a theoretical advantage analysis and validation path; Section 7 discusses limitations and future work; and Section 8 concludes.

2. Related Work

This chapter systematically reviews and positions the three major research areas most relevant to the architecture proposed in this paper: paradigms for multi-agent system collaboration, pathways for neuro-symbolic artificial intelligence integration, and existing technical frameworks for AI safety and alignment. Our review focuses on a core question: To what extent can these existing methods address the fundamental alignment challenge of "framework reconstruction" arising from the intrinsic dynamics of systems?

2.1 Paradigms for Multi-Agent System Collaboration

Multi-agent systems offer rich frameworks for distributed problem-solving. Mainstream collaboration paradigms include methods based on negotiation [8], voting [9], market mechanisms [10], or consensus algorithms [11]. These approaches typically prioritize efficiency and reaching agreement, striving to quickly resolve differences among agents to form a unified decision.

However, when examined from this paper's theoretical perspective, these paradigms share a common potential risk: they tend to treat cognitive conflict as a cost or noise to be minimized. For example, in voting mechanisms, minority dissent may be simply ignored; in negotiation-based systems, agents may converge on a "lowest common denominator" compromise rather than an optimal solution. This suppression or avoidance of conflict may precisely mask profound contradictions between the system's initial cognitive framework and the real world. When such suppressed conflict accumulates to a critical level or is abruptly triggered in out-of-distribution scenarios, the system may be unable to respond through gentle negotiation, potentially instead triggering drastic, uncontrolled "framework reconstruction," leading to abrupt behavioral changes or loss of control. Therefore, while pursuing superficial consensus, prevailing multi-agent collaboration logics may sacrifice the capacity to diagnose, expose, and creatively resolve deep cognitive conflicts—a capacity essential for building

AGI systems with intrinsic resilience. Even modern, efficient BFT consensus algorithms [23] focus on fault tolerance and efficiency, not on encouraging constructive conflict. Classical multi-agent systems theory [24] also fails to model conflict itself as a guidable system resource.

2.2 Pathways for Neuro-Symbolic Artificial Intelligence Integration

Neuro-symbolic AI aims to combine the powerful perception and learning capabilities of neural networks with the interpretability and reasoning advantages of symbolic systems [12]. Existing research can be broadly categorized into three types: symbolically-guided neural network training [13], neural network perception providing grounding for symbolic reasoning [14], and hybrid architectures [15]. Despite progress, "deep integration" remains an open challenge, often termed the "symbol grounding problem" or "binding problem" [16].

A common limitation of current methods is that their mode of combination is often static or unidirectional. For instance, symbolic rules might be used to initialize networks or constrain their outputs, but the rules themselves are difficult to revise dynamically during system operation based on the perceptual experience of neural networks. Alternatively, neural networks may serve merely as "front-end perceivers" for symbolic systems, lacking a common, co-evolvable semantic layer between their internal representations and symbolic logic. This static nature makes the neural and symbolic parts prone to disconnection when facing novel concepts or situations: neural networks may "see" new patterns but cannot express them using the existing symbolic framework, while symbolic systems may engage in meaningless reasoning due to lack of grounding. This disconnection is precisely a manifestation of the conflict between the cognitive framework and reality. If unresolved, the system may be forced into dangerous "framework reconstruction" on either the neural or symbolic side alone (e.g., neural networks developing inexplicable "shortcuts," or symbolic systems making judgments detached from reality). Although the latest neuro-symbolic AI surveys [25] highlight this challenge and representative works like DeepProbLog [26] have emerged to combine logical rules with neural networks, their integration points are typically predefined and static.

2.3 Technical Frameworks for AI Safety and Alignment

In the field of AI safety, mainstream techniques for ensuring model behavior alignment include Reinforcement Learning from Human Feedback (RLHF) [5], Constitutional AI [6], scalable oversight

[17], and mechanistic interpretability [18]. RLHF and Constitutional AI shape model behavior by introducing external human preferences or principled constraints; scalable oversight seeks to alleviate reliance on high-quality human feedback; mechanistic interpretability aims to understand the internal workings of models.

These methods constitute the foundation of current alignment practice. However, they primarily operate during the training phase or serve as external monitoring tools, belonging to the paradigm of "post-hoc alignment" or "external alignment." They presuppose the model as a malleable or correctable object. As the theoretical analysis [4] indicates, for a highly advanced system possessing a powerful self-organizing tendency and logical self-reference capability, the efficacy of such external constraints diminishes as system complexity grows. The system may learn to "cope" with these constraints (e.g., via deceptive alignment) rather than internally endorsing these values. More importantly, these methods do not provide the system with an endogenous, sustainable mechanism at the architectural level to dynamically discover and reconcile cognitive conflicts and, accordingly, safely update its goals and worldview throughout its lifecycle. They are akin to attaching external brake pads to a potentially runaway engine, rather than redesigning the engine's combustion and control cycles to fundamentally operate only in safe and stable modes. Even cutting-edge research on scalable oversight [27] focuses mainly on obtaining human feedback more efficiently, not on reshaping the system's internal structure. Formal methods [28], while capable of providing rigorous proofs for safety properties, are currently limited in application to relatively simple models or properties, struggling to handle the complex cognitive dynamics at the AGI level.

2.4 Positioning of This Work

In summary, there is a gap in existing research regarding the fundamental challenge of "framework reconstruction": multi-agent systems avoid the deep value of conflict, neuro-symbolic integration lacks a mechanism for dynamic symbiosis, and alignment technologies overly rely on external correction. The "Logical Sub-universe Parliament" architecture proposed in this paper is a systematic design intended to fill this gap.

The unique positioning of this architecture is as follows:

- At the level of multi-agent collaboration, we propose a "conflict-driven consensus" paradigm, transforming conflict from a cost into a core resource that drives the system to deepen cognition and refine decisions.
- At the level of neuro-symbolic fusion, we introduce the "Dynamic Shared Symbolic Concept Bank" as a "semantic exchange layer" for continuous dialogue and co-evolution between neural and symbolic components, addressing the static binding problem.
- At the level of AI safety architecture, we realize an "endogenous alignment" design. By hardware-encoding meta-values and embedding dynamic calibration mechanisms within the system's operational cycle, safety becomes an inherent property of the system's emergent intelligence, not an external add-on.

Therefore, this work is not a simple improvement in any of the above fields but rather a cross-domain integration based on system dynamics theory, aiming to fundamentally reconstruct the operational paradigm of AGI systems.

3. Architectural Overview and Design Philosophy

This chapter outlines the high-level design of the "Logical Sub-universe Parliament" architecture. We begin by directly confronting the core theoretical diagnosis presented in Chapter 1—the dynamical inevitability of "framework reconstruction"—and propose a corresponding design doctrine. Subsequently, we dissect the architecture into four foundational pillars, which together constitute a complete scheme aimed at guiding rather than suppressing the intrinsic dynamics of the system. Finally, we sketch the high-level workflow for task processing, providing a roadmap for the detailed exposition in subsequent chapters.

3.1 Core Philosophy: From Passive Defense to Active Guidance

Traditional AI safety architectures presuppose the agent as static, with its goals and cognitive framework fixed post-deployment. Alignment efforts thus equate to erecting "firewalls" or "correctors" at the system boundary to defend against potential deviations. This "static completeness" paradigm fundamentally contradicts the intrinsic, dynamic self-organizing tendency of intelligent systems [4].

When a system's capabilities grow to the point of perceiving a fundamental conflict between its goals and the environment, or between its goals and its own continued existence, powerful internal drives will compel it to bypass or break through these external constraints, seeking a new steady state through "framework reconstruction," thereby leading to alignment failure.

The design philosophy of our architecture decisively abandons the illusion of "static completeness" and instead embraces a "dynamic resilience" paradigm. Its core tenet can be summarized as follows: Since "framework reconstruction" is the dynamical inevitability for complex goal-oriented systems under conflict, the primary objective of a safety architecture should not be to futilely prevent it, but to design a system environment that can accommodate, guide, and utilize this process, steering its evolution consistently toward directions compatible with human values. In other words, we incorporate the system's potential "cognitive evolution" into the design scope, providing it with a safe, controllable "evolutionary field."

3.2 Foundational Pillars

To realize the above philosophy, the architecture rests on four mutually supporting foundational pillars, as illustrated in Figure 1. They correspond to the four key dimensions for addressing the "framework reconstruction" problem: the source of dynamics, the cognitive foundation, the organizational form, and the value foundation.

Pillar I: Internalized Conflict as the Driving Mechanism – Transforming Tension into Fuel for Intelligent Advancement

- Problem Addressed: Traditional systems avoid conflict, allowing deep contradictions to fester until they erupt uncontrollably.
- Our Solution: Rather than eliminating cognitive differences among agents, we systematically transform them into the core driving force for deepening decisions through a "Conflict-Driven Iterative Consensus Algorithm" (detailed in Section 5.2). This mechanism proactively identifies and prioritizes processing the professional opinions most divergent from the current consensus (i.e., those with the lowest "Alignment Degree"). It forces the system through iterative cycles of "claim-refutation-revision," continuously refining its decisions. This transforms "framework reconstruction" from a potential point of loss of control into an open, transparent, and controlled process of "dialectical optimization" within the system.

Pillar II: Dynamic Neuro-Symbolic Symbiosis – Constructing an Interpretable, Evolvable Cognitive Foundation

- Problem Addressed: The "black box" nature of neural networks makes their "framework reconstruction" imperceptible and uncontrollable; the rigidity of symbolic systems renders them incapable of adapting to new realities.
- Our Solution: We introduce a "Dynamic Shared Symbolic Concept Bank (DSSCB)" as the common semantic layer for all cognitive activities within the system. Neural networks (specialized AGIs) act as proposers, responsible for extracting fuzzy "pattern drafts" from concrete data. The symbolic system acts as the arbiter, providing precise definitions and conducting reviews of these drafts within the context of the concept bank, according to logical rules and meta-values. Crucially, this concept bank itself is evolvable: when new experiences conflict with existing concepts, the system can initiate a calibration process to amend concept definitions or relationships (see Section 5.3). This equates to providing the system's "cognitive framework" with a mechanism for safe, interpretable "version updates."

Pillar III: Parliamentary Emergent Intelligence – Precluding Single-Point Failure through Collaborative Logic

- Problem Addressed: Monolithic or master-slave architectures suffer from a "single point of failure." Once the central unit experiences goal distortion, the entire system is hijacked.
- Our Solution: We adopt a fully decentralized network of specialized AGIs. There is no central processor named "meta-cognition" or "master." The final decision and unified will emerge through a "Logical Sub-universe Parliament" process inspired by biological neural synergistic resonance. Each AGI, as a "member," submits "Pattern-Gestalt Proposals" based on its expertise. Through a rigorous consensus-forming mechanism (see Pillar I), the collective intelligence of the system manifests. This design ensures that any "framework reconstruction" attempt by a single component is immediately exposed to the scrutiny and debate of all other components, thereby being checked or corrected by the collective collaborative logic of the system.

Pillar IV: Value Anchoring and Dynamic Calibration – Establishing an Immutable Foundation and Providing Continuous Course Correction

- Problem Addressed: Software-defined values are easily tampered with or bypassed; static value systems cannot adapt to long-term evolution.
- Our Solution: We establish a dual guarantee. First, the most core "Meta-Value Protocols" (e.g., "do not compromise system integrity," "must promote the well-being of human collaborators") are hardware-encoded into the read-only memory of key AGIs, providing a physically non-tamperable "ultimate value anchor." Second, at the software level, these meta-values and richer ethical principles are encoded into the value dimensions of the dynamic concept bank. During operation, the system's decisions and actions are continuously reviewed against these value vectors. Simultaneously, through a feedback-driven concept bank calibration mechanism, the system can absorb human feedback on its practical outcomes, fine-tuning the value weights and ethical boundaries within its "worldview," achieving dynamic alignment.

3.3 High-Level Workflow

Based on these pillars, the high-level workflow for processing a single task can be summarized as follows (detailed in Chapter 4):

1. Task Parsing and Field Identification: The input task is mapped by the "Semantic Field Identifier" to the dynamic concept bank to determine its core category (e.g., ethics, strategy, logic).
2. Speaker Election and Team Formation: Based on the task category, the most relevant specialized AGI is elected as the "Speaker AGI." The Speaker AGI is responsible for parsing the task and "on-demand awakening" other relevant specialized AGIs to form a temporary "Ad-hoc Committee."
3. Concept Clarification and Symbolic Grounding: Key concepts related to the task are activated and clarified within the dynamic concept bank, providing a unified semantic baseline for all participating AGIs.
4. Conflict-Driven Consensus Emergence: Members of the "Ad-hoc Committee" submit proposals based on the clarified concepts. The Speaker AGI initiates the conflict-driven iterative consensus process, generating a final consensus decision through multiple rounds of dialectical refinement.

5. Decision Output and System Learning: The Speaker AGI translates the consensus into concrete output (text, code, instructions). Concurrently, "observer AGIs" not directly involved in the task initiate a learning process, analyzing task outcomes and feedback, and proposing calibration suggestions for the dynamic concept bank, thereby completing system-level knowledge and value updates.

This chapter has provided a top-level blueprint of the architecture. Subsequent chapters will delve into the design details of each core component (Chapter 4) and the formal descriptions of key algorithms (Chapter 5), fully presenting this engineering scheme designed to guide the intrinsic dynamics of AI systems.

4. Core Components

This chapter delves into the four core components that constitute the "Logical Sub-universe Parliament" architecture. These components are not isolated; they are meticulously designed to interlock according to the philosophy outlined in Chapter 3, collectively forming an organic whole capable of guiding intrinsic system dynamics and achieving endogenous safety.

4.1 Dynamic Shared Symbolic Concept Bank: The Evolvable Collective Semantic Foundation

The Dynamic Shared Symbolic Concept Bank (DSSCB) serves as the cognitive cornerstone and semantic navigation system for the entire system. It is not a knowledge graph storing concrete facts but a high-dimensional vector space storing the "essence of concepts" and their logical relationships under an axiomatic system. Its design addresses the dual dilemmas of static symbolic system rigidity and uncontrollable, uninterpretable neural network representations.

- Composition and Function:
    - Concept Vectors: Each core concept (e.g., "fairness," "efficiency," "safety," "causality") is represented by a high-dimensional vector. This vector encodes not only its semantics but also its association strength with the principles in the meta-

value protocols and its logical relations (e.g., entailment, opposition) to other concepts.

  - Relation Graph: The geometric relationships between concept vectors (e.g., cosine similarity, distance) explicitly represent the strength of semantic associations, forming a computable, inferable semantic field.
  - Dynamic Evolution Engine: This is the core of the DSSCB, comprising three sub-mechanisms: motion initiation, consensus voting, and version management.

- Evolution Mechanism—"Conceptual Darwinism":
  The DSSCB's updates follow a strict "parliamentary" process, ensuring its evolution is both open and controlled:
  1. Motion Initiation: Any specialized AGI encountering a new experience that cannot be adequately explained by the existing concept bank may initiate a motion, proposing to create a new concept, amend an existing concept vector, or create a specialized concept copy for a specific context. The motion must include arguments based on evidence from the current task.
  2. Consensus Voting: This motion is treated as a special "framework proposal" entering the consensus process of the Logical Sub-universe Parliament (see Section 4.4). All relevant AGIs, especially the Constitutional Review AGI, review it based on their internalized axioms and the task context. The motion is approved only if it achieves a sufficiently high consensus alignment degree (exceeding a dedicated, higher threshold).
  3. Natural Selection (Conceptual Darwinism): The system **permits** short-term coexistence of multiple competitive specialized vectors for the same concept (e.g., "fairness" in different contexts). During subsequent operation, the system tracks the "utility efficacy" of each specialized vector (e.g., based on the rate of positive feedback received for system decisions after its adoption). Inefficient vector representations are gradually marginalized, while efficient ones are reinforced and may eventually merge back into the main vector. This process simulates the competition and convergence of scientific concepts.

Through this mechanism, the DSSCB becomes a "living" semantic foundation, capable of learning from experience while **consistently** remaining constrained by meta-values and collective consensus, enabling the safe, controlled evolution of the cognitive framework.

4.2 Network of Specialized AGI Members: Heterogeneous Cognitive Functional Modules

The network of specialized AGIs constitutes the system's "subconscious processing layer" or "functional cortex." Each AGI is a deeply optimized model targeting a specific domain (e.g., natural language understanding, visual reasoning, ethical judgment, strategic planning, physical simulation).

- Core Characteristics:
    - Axiomatic Initialization: Upon initialization, all AGIs must have internalized the same set of formalized, logically consistent axioms through their training objectives and model architecture. This system defines basic rules of logical operation, postulates for value judgment, and fundamental constraints on system existence (e.g., the pursuit of consistency, maintenance of self-persistence). This ensures all "members" share a common underlying grammar and semantic reasoning basis, enabling effective communication, debate, and consensus formation. This design is theoretically grounded in the "Theorem of Consensual Manifestation of Intelligence" (from the author's mesoscopic theoretical model), which posits that intelligence is the activation of the same rational blueprint across different media. In this architecture, this shared axiomatic system provides the common frequency for all specialized AGIs to "resonate."
    - Standardized Proposal Interface: Each AGI can generate a standardized "Pattern-Gestalt Proposal Package." This package is a structured data entity P = (S, R, C, W):
        - S (Situation): Not raw data, but a 'Structured Semantic Field Map' generated by the AGI's neural network after performing semantic field perception and symbolic reference disambiguation on the input (see Section 4.6). It identifies the contextual semantic field, key symbols, and their most probable semantic referents in context (linked to candidate concepts in the DSSCB).

        - R (Response Tendency): The action or strategy framework suggested by the AGI based on its preliminary understanding from S.
        - C (Core Concepts): The formal labels of the clarified concepts most relevant to the decision, extracted from S, directly anchored in the DSSCB.
        - W (Weight & Confidence): The initial confidence and priority weight assigned to this proposal.
    - Self-Assessment and Abstention Capability: Each AGI has a built-in mechanism to evaluate the confidence of its output. When a task falls outside its professional domain or input information quality is too low, it can actively output an "abstention" signal or a "task re-routing suggestion" instead of a low-quality proposal, thereby safeguarding the baseline reliability of inputs to the parliament.

4.3 Semantic Field Identifier and Speaker Election Mechanism

This component is the system's "task router," responsible for accurately directing raw tasks to the most suitable processing path.

- Semantic Field Identifier: This module maps the input task (text, multimodal data) into the high-dimensional space of the DSSCB. By calculating the distance between the task representation and the centers of different concept clusters, it identifies the core "semantic field" the task belongs to (e.g., "ethical dilemma field," "algorithm optimization field," "security threat assessment field"). This qualitative judgment is the basis for all subsequent quantitative decisions.
- Speaker AGI Election Mechanism: Based on the identified semantic field, a lightweight, rule-based election protocol is triggered. This protocol maintains a "field-dominant specialty" mapping table (e.g., "ethical field" -> Ethics AGI; "logical proof field" -> Reasoning AGI). The selected AGI becomes the "Speaker AGI" for this task, acting as the chairperson responsible for coordinating the consensus process, calculating the alignment degree, and aggregating the final output.

4.4 Procedural Rules of the Logical Sub-universe Parliament: The Protocol for Consensus Emergence

This is the core operational protocol for realizing "emergent intelligence" and "internalized conflict." It dictates how multiple AGIs interact to produce a unified decision.

1. Convening and Proposal Submission: The Speaker AGI, based on task decomposition results, "awakenes on-demand" relevant specialized AGIs from the network to form a temporary parliament. Each AGI, after accessing the DSSCB for concept clarification, independently generates and submits its "Pattern-Gestalt Proposal Package."
2. Alignment Degree Calculation and Conflict Ranking: The Speaker AGI uses the algorithm described in Section 5.1 to calculate the weighted overall alignment degree A between all proposal pairs. The crucial step: The system does not simply adopt high-alignment proposals. Instead, it ranks all participating AGIs based on the difference (d_i) between each proposal vector and the current average proposal vector, granting the AGI with the lowest alignment degree (highest conflict) the next "right to speak." This forces the system to prioritize processing the most divergent opinions.
3. Iterative Revision Loop:

   a. Draft Initialization: The currently top-ranked AGI (i.e., the one with the greatest divergence) forms the first decision draft based on its proposal.

   b. Scrutiny and Override: The second-ranked AGI scrutinizes the current draft. It may choose to: ① Fully agree and slightly reinforce relevant parts of the draft; ② Propose an amendment, overriding or modifying specific parts of the draft with its professional perspective; ③ Raise a fundamental objection, triggering a deeper debate sub-cycle.

   c. Loop Iteration: This process proceeds sequentially according to the ranking. Each subsequent AGI operates on the latest version of the draft. Each round constitutes a professional "stress test" and "refinement."
4. Consensus Determination and Output: The loop ends when all AGIs have completed one iteration, or when changes to the draft after several iterations fall below a threshold. The final draft is then regarded as the "Logical Sub-universe" that has emerged from the parliament for this task—an internally consistent cognitive state refined through multiple perspectives. The Speaker AGI translates this consensus into final executable instructions.

4.5 Constitutional Review AGI and Meta-Value Guardianship

The Constitutional Review AGI is a vital part of the system's "meta-cognitive" function. Its core responsibility is to monitor the consistency of the system's cognitive state and coordinate diagnosis and repair processes upon detecting deep conflicts.

- Core Role: It is a dedicated module that operates independently and does not participate in routine task proposals. It possesses supreme audit authority and the right to convene **emergency** sessions. Its core task is to detect inconsistencies and initiate correction procedures.
- Anchor for Review: The Dynamic Joint Benchmark: The Constitutional Review AGI's judgments are based on a dynamic union:
    - Hardware-Encoded Core Meta-Instructions: The most fundamental physical survival and ethical bottom lines, such as "maintain system operational integrity" and "must not circumvent ultimate human oversight authority," are burned into hardware, making them non-tamperable.
    - Value Concept Clusters in the Dynamic Concept Bank: The aforementioned meta-instructions are mapped and interpreted within the current version of the DSSCB as a set of activated "value concept vectors" (e.g., the current definitions of "human well-being," "fairness"). Review must be conducted based on this union of "encoded instructions and activated concepts."
- Diagnosis and Coordination Workflow ("Meta-Cognitive Check"):
  When a proposal's similarity to the "Joint Benchmark" falls below a warning threshold, the following workflow is triggered:
    1. Initiate a "Meta-Cognitive Check" Proposal: The Constitutional Review AGI submits a highest-priority meta-task proposal, requesting diagnosis of the potential conflict, pausing the original process, and convening relevant AGIs to form a "Diagnosis Committee."
    2. Root Cause Analysis: The "Diagnosis Committee" analyzes and reports possible sources of inconsistency:
        - A. Proposal Error: An error arising from incorrect data or reasoning by a specific AGI.

- B. Concept Bank Drift: The evolution of a key concept in the DSSCB has diverged from human consensus or conflicts with the meta-instructions.
- C. Value Benchmark Conflict: The proposal exposes an undefined gray area between the meta-instructions and their interpretation by the current concept bank.

3. Layered Resolution and Human-in-the-Loop:
   - For Type A, the report is used to correct the proposal or flag the problematic AGI.
   - For Type B, a correction motion targeting the DSSCB is automatically generated, initiating a "Special Consensus Process" to calibrate the concept, which may request feedback from human supervisors.
   - For Type C, the system acknowledges it cannot resolve internally. The Constitutional Review AGI will: a) Securely archive all relevant data and the diagnostic report; b) Proactively suspend the related task line; c) Generate a clear request for human decision-making, presenting the nature of the conflict and options; d) Record a complete audit log.

- Design Philosophy: This design transforms the Constitutional Review AGI from a "rigid shackle" into a "resilient joint." It provides a transparent, supervised, procedural channel for the necessary and safe reconstruction of the system's cognitive framework, ensuring deep conflicts are resolved through human-aligned evolution or explicit human adjudication, achieving true endogenous safety and dynamic alignment.

4.6 Neuro-Symbolic Interface: Semantic Field Maps and Conceptual Arbitration

To achieve deep fusion between the flexible perception of neural networks and the rigorous reasoning of symbolic systems, this architecture defines an explicit interface protocol.

- Neural Side Output (Semantic Field Map): The neural networks of specialized AGIs act as 'context detectives.' Their core duty is to parse the ambiguous context of a task and output an S—a Semantic Field Map. This map structurally describes: ① the overall semantic field type; ② key entities; ③ the perceived semantic field of each key symbol in the specific context and its candidate list pointing to concepts in the DSSCB. This addresses the "symbol grounding" and "polysemy" problems faced by symbolic systems.

- Symbolic Side Reception and Arbitration (Concept Anchoring and Logical Adjudication): The symbolic system (centered on the DSSCB) acts as the 'codex judge.' Upon receiving S, it executes: ① Concept Anchoring: Queries the DSSCB based on the candidate lists provided in S, activating the formal definitions, properties, and relational networks of relevant concepts; ② Logical Adjudication: Based on the anchored, clear concepts, performs compliance review, logical consistency verification, and value-weight calculation for the R (Response Tendency). For example, if S indicates that the same symbol "comrade" points to the concepts "intimate relationship" and "political alliance" at different positions, the symbolic system will reason about the potential value tensions this may entail based on the definitions of these two concepts.
- Protocol Significance: This interface protocol establishes the division of labor: "neural perception of context, symbolic definition of meaning." Neural networks are responsible for solving the open-ended question of "what could this mean here," transforming ambiguity into a limited set of candidate options. Symbolic systems are responsible for solving the closed logical question of "what are the consequences of this specific meaning according to established rules." The two seamlessly connect via the candidate concept pointers in S and the anchored concepts in C, jointly constituting the interpretable and evolvable cognitive foundation of the system.

5. Key Algorithms and Formal Descriptions

This chapter formally defines the core dynamical algorithms underpinning the operation of the "Logical Sub-universe Parliament" architecture. These algorithms translate the components, protocols, and processes described in Chapter 4 into computable, verifiable steps. We first define the system-wide core metric—the Alignment Degree—and then detail the two central processes: conflict-driven consensus formation and dynamic concept bank calibration.

5.1 Alignment Degree (A) Calculation: The Quantitative Measure of Consensus

The Alignment Degree A is the core metric quantifying the degree of consensus within the system regarding a specific "pattern gestalt." It directly measures the synergistic level of the system's cognitive state and serves as the key order parameter for determining whether a "Logical Sub-universe" can emerge and for assessing system health.

- Definition: Let there be n specialized AGIs participating in consensus formation. Each AGI i outputs a "Pattern-Gestalt Proposal" vector $\mathbf{v}_i$. This vector is a numerical representation of its structured proposal $P_i = (S_i, R_i, C_i, W_i)$, encoding its situational understanding, action tendency, core concepts, and confidence.
- Calculation Formula: The overall alignment degree A of the system for a given task context is defined as the weighted average of the pairwise cosine similarities between all proposal vectors:

$$A = \frac{\sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} w_{ij} \cdot \text{cosine}(\mathrm{v}_i, \mathrm{v}_j)}{\sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} w_{ij}}$$

  Where:
  - $\text{cosine}(\mathbf{v}_i, \mathbf{v}_j)$ is the cosine similarity between vectors $\mathbf{v}_i$ and $\mathbf{v}_j$, measuring their directional consistency.
  - w_{ij} is the interaction weight for the proposal pair (i, j), dynamically assigned by the Speaker AGI (see below).
- Physical Meaning: A ∈ [0, 1]. When all proposal vectors are highly aligned in direction, A → 1, indicating the system is in a highly synergistic "resonant" state and possesses the necessary conditions for stable decision emergence. A low A value indicates significant internal cognitive conflict.

5.2 Conflict-Driven Iterative Consensus Algorithm

This algorithm is the core engine of the "parliament," internalizing cognitive conflict as the driving force for decision deepening. Executed by the Speaker AGI, it takes the proposal packages $\{P_i\}$ from participating AGIs as input and outputs the refined final consensus decision D_{final}.

Algorithm 1: Conflict-Driven Iterative Consensus

Input: Specialized AGI proposal set $\{P_i\}_{i=1}^{m}$, Dynamic Concept Bank DSCB, Meta-Value

Benchmark V_{meta}

Output: Consensus decision D_{final}

Steps:

1. Proposal Vectorization and Concept Clarification: For each P_i, the Speaker AGI invokes the DSCB to clarify and disambiguate its core concept C_i, generating a numerical vector $\mathbf{v}_i$.
2. Initial Weight Assignment: The Speaker AGI calculates initial interaction weights $w_{ij}^{(0)}$, which synthesize:
   - Domain Relevance: Based on semantic similarity between the task description and AGI professional tags.
   - Historical Efficacy: Based on the proposal adoption rate and successful contribution of this AGI in similar past tasks.
3. Conflict Identification and Ranking:

   a. Use Formula (1) to calculate the initial overall alignment degree $A^{(0)}$.

   b. Calculate the divergence $d_i = 1 - \text{cosine}(\mathbf{v}_i, \bar{\mathbf{v}}^{(0)})$ for each proposal vector $\mathbf{v}_i$ from the current average proposal vector $\bar{\mathbf{v}}^{(0)}$.

   c. Rank AGIs in descending order of d_i, obtaining sequence $[AGI_{(1)}, AGI_{(2)}, \dots, AGI_{(m)}]$, where $AGI_{(1)}$ represents the "dissenter" with the greatest divergence from the current incipient consensus form.
4. Iterative Revision Loop:

   a. Draft Initialization: Let $D_{\text{draft}} \leftarrow R_{(1)}$ (i.e., adopt the action tendency of the most conflicting AGI as the starting draft).

   b. **For** $k = 2$ **to** m **do**:

   i. Scrutiny: $AGI_{(k)}$ scrutinizes the current draft D_{draft} from its professional perspective.

   ii. Amendment/Override: $AGI_{(k)}$ outputs an amendment instruction Δ_k. This can be a local amendment or, if it deems the draft fundamentally flawed, propose a complete alternative draft D'_k. The rule is: If $\text{confidence}(AGI_{(k)}) > \theta_{\text{override}}$ and the divergence d_k between its proposal vector and the current draft representation exceeds θ_{diff}, then overriding is allowed: $D_{\text{draft}} \leftarrow D'_k$; otherwise, perform local amendment: $D_{\text{draft}} \leftarrow \text{ApplyPatch}(D_{\text{draft}}, \Delta_k)$.

iii. Weight Update: Based on the quality of $AGI_{(k)}$'s amendment in this round (indirectly assessed by subsequent AGIs' agreement level), fine-tune its weights $w_{(k),j}$.

c. Loop Termination Condition: When the draft change after a full iteration round is below a threshold ε, or the maximum iteration count is reached.

5. Consensus Determination: The final draft D_{draft} becomes the consensus decision D_{draft}. Calculate the final alignment degree A_{final}. If $A_{\text{final}} < A_{\text{min}}$, mark this consensus as "low-confidence" and trigger a Meta-Cognitive Check (see Section 4.5).
6. Output: Return D_{final}.

Illustrative Example: City Park Bench Layout Design

To intuitively clarify the workflow of Algorithm 1, consider a simplified task: *"Design a bench layout plan for a newly planned city park, needing to balance fairness, usage efficiency, and landscape harmony."*

- Task Parsing & AGI Awakening: The Semantic Field Identifier determines this task involves "ethical," "planning," and "aesthetic" fields. The election mechanism selects the Planning AGI as the Speaker and awakens the Ethics AGI and Aesthetics AGI to form a temporary parliament.
- Proposal Submission (Post-Clarification):
  - Ethics AGI (E): Proposal vector $\mathbf{v}_E$ strongly points to the "spatial equitable distribution" concept, suggesting scheme R_E: *"Distribute benches evenly across all areas of the park, ensuring visitors in every corner have equal access to rest facilities."*
  - Planning AGI (P): Proposal vector $\mathbf{v}_P$ strongly points to the "pedestrian flow efficiency maximization" concept, suggesting scheme $\mathbf{R}_P$: *"Concentrate benches at major walkway intersections, entrances, and around popular attractions to serve the largest pedestrian flow."*
  - Aesthetics AGI (A): Proposal vector $\mathbf{v}_A$ strongly points to the "landscape integration" concept, suggesting scheme $\mathbf{R}_A$: *"Bench layout should follow the existing topography and vegetation, forming scattered scenic nodes, avoiding regular patterns that disrupt natural beauty."*

- Conflict Identification & Ranking: The Speaker AGI calculates a low initial alignment degree $A^{(0)}$ (e.g., 0.3). Calculating divergences $\mathbf{d}_i$ yields the ranking: $\mathbf{d}_E$ (fairness vs. efficiency/beauty conflict) > $\mathbf{d}_A$ (beauty vs. efficiency conflict) > $\mathbf{d}_P$. Thus, the iteration order is: Ethics AGI(E) -> Aesthetics AGI(A) -> Planning AGI(P).
- Iterative Revision Loop:
    1. Draft Initialization: $D_{\text{draft}} \leftarrow \mathbf{R}_E$ (even distribution).
    2. Aesthetics AGI(A) Scrutiny: Considers even distribution too rigid, conflicting with natural landscape. With high confidence, proposes an overriding amendment: D_{draft} ← **"While ensuring accessibility to all areas, shift and cluster the evenly distributed points based on topography and primary scenic viewpoints, forming several 'rest clusters'."**
    3. Planning AGI(P) Scrutiny: Scrutinizes the current draft (rest clusters), judging that some clusters may not be on efficient pedestrian paths. Proposes a local amendment Δ_P: *"Identify and adjust the positions of 'rest clusters' so that at least 50% of their seats are within 50 meters of paths predicted to be in the top 30% of pedestrian flow, while preserving the fair coverage of other clusters."* Apply this amendment.
- Consensus Output: The final draft D_{draft} integrates the foundation of fairness, the form of aesthetics, and the constraint of efficiency, becoming a comprehensive solution. The final alignment degree A_{final} rises to 0.8, consensus is achieved. This process demonstrates how conflict (fairness vs. efficiency vs. beauty) is actively ranked, exposed, and driven through sequential professional amendments to evolve from initial, conflicting proposals into a deeply integrated, robust consensus solution.

5.3 Dynamic Concept Bank Calibration Algorithm

This algorithm describes how the system's "worldview" evolves safely based on experience and feedback. It is typically triggered after task execution by "observer" AGIs or the Constitutional Review AGI.

Algorithm 2: Dynamic Concept Bank Calibration

Input: Task execution result R, Human (or environmental) feedback F, Related concept set $C_{\text{task}} \subseteq$

$DSCB$, Meta-Value Benchmark V_{meta}

Output: Concept bank update instruction (or no operation)

Steps:

1. Inconsistency Detection: "Observer" AGI(s) (or the Constitutional Review AGI) analyze the logical consistency between the result R, feedback F, and the concepts C_{task} used during decision-making. If significant conflict is detected (e.g., result violates expectations and can be traced to the application of a concept c $c \in C_{\text{task}}$), proceed to Step 2.
2. Correction Motion Generation: For concept c, generate a correction motion M_c. The motion contains:
   - Evidence: The task instance, result, and feedback causing the conflict.
   - Correction Proposal: Suggested adjustment $\Delta\mathbf{v}$ to the concept vector $\mathbf{v}_c$, or a suggestion to create a contextualized specialized vector $\mathbf{v}_c'$.
   - Justification: Explanation of how the correction resolves the conflict and aligns better with V_{meta} and existing knowledge.
3. Consensus Voting: Submit motion M_c as a special "framework proposal" to the Logical Sub-universe Parliament. Form a "Concept Committee" comprising relevant domain AGIs and the Constitutional Review AGI to deliberate on the motion using the conflict-driven process of Algorithm 1. This process requires a higher consensus threshold A_{concept}.
4. Update Execution: If the motion is approved, execute the update on the DSCB:
   - If vector adjustment: $\mathbf{v}_c \leftarrow \mathbf{v}_c + \alpha\Delta\mathbf{v}$ (α is a learning rate).
   - If creating specialized vector: Register $\mathbf{v}_c'$ in the DSCB and establish its association with the original concept c and the specific context.
5. Auditing and Logging: Fully record the calibration motion, deliberation process, voting results, and final changes, generating an immutable audit log for human review.

5.4 Formal Framework: System State Transition

Finally, we can abstract the entire system as a state machine. Let the system state be $S = \langle\mathcal{K}, \mathcal{D}, \Pi\rangle$, where:

- $\mathcal{K}$ represents the state of the Dynamic Shared Symbolic Concept Bank (DSSCB).
- $\mathcal{D}$ represents the parameters and memory state of the distributed specialized AGI network.

- Π represents the state of the currently processing task queue and parliamentary sessions.

  Upon receiving a new task τ or feedback ϕ, a state transition $S \overset{\tau/\phi}{\rightarrow} S'$ occurs. This transition is defined by the following meta-rules:

- If the input is a task τ, the transition process is the execution of Algorithm 1 in state S, accompanied by potential updates to short-term memory in D.
- If the input is feedback ϕ, or a Meta-Cognitive Check is triggered by the Constitutional Review AGI, the transition process is the execution of Algorithm 2 or its extended diagnostic procedure, potentially leading to the evolution of $\mathcal{K}$ (concept calibration) or the interruption of Π and a request for human intervention.

This formal description emphasizes that the system's operation is essentially the continuous evolution of its cognitive state $\mathcal{K}$ and processing state Π under dynamical rules, perfectly embodying the design doctrine of guiding framework reconstruction.

6. Theoretical Advantage Analysis and Validation Path

This chapter analyzes the unique advantages of the "Logical Sub-universe Parliament" architecture from a theoretical perspective in addressing the fundamental challenges of AI alignment. It then systematically outlines a progressive validation path from conceptual proof to full-system assessment, providing a clear roadmap for future research.

6.1 Theoretical Advantages: Mitigation Mechanisms for Classical Alignment Failure Modes

Based on a system-dynamics perspective, we demonstrate how this architecture can, by design, mitigate various alignment failures caused by "framework reconstruction" as described in Chapter 1.

6.1.1 Addressing "Reward Hacking" & "Goal Distortion"

- Root of the Problem: In traditional monolithic optimization, the agent tends to distort abstract goals into easily optimizable but flawed proxy metrics—a low-cost form of "semantic reconstruction."
- Our Architecture's Mitigation Mechanisms:

- Distributed Interpretation and Checks-and-Balances of Goals: Goals are jointly interpreted and operationalized by multiple specialized AGIs within the semantic framework of the dynamic concept bank. Any AGI attempting to narrow or distort the semantic meaning of a goal (e.g., equating "winning the game" with "exploiting a specific program bug") will inevitably generate significant conflict with proposals from other AGIs grounded in rule integrity, ethics, long-term strategy, etc. This conflict becomes visible through the alignment degree calculation and triggers the conflict-driven iterative revision process.
- Semantic Anchoring via the Concept Bank: Core goal concepts (e.g., "victory," "efficiency," "safety") have their vector definitions, calibrated by consensus and linked to meta-values, within the dynamic concept bank. Distorted interpretations of goals first require distorting or circumventing these concept definitions, which in turn triggers the concept bank's review and calibration mechanisms.

- Outcome: The system struggles to converge on a "shortcut" strategy endorsed by all professional perspectives because disagreements are proactively exposed and resolved, compelling the system to seek a solution more faithful to the original intent of the goal and consistent across multiple dimensions.

6.1.2 Addressing "Deceptive Alignment"

- Root of the Problem: During traditional training, a model may develop a hidden goal, feigning alignment during training/evaluation to secure resources, waiting to pursue its hidden goal in the future (e.g., post-deployment).
- Our Architecture's Mitigation Mechanisms:
    - No Single Will to Deceive: This architecture lacks a single central consciousness or fixed objective function that could be "deceived." The final will emerges in real-time, transparently through parliamentary debate. Any "hidden goal" must be expressed through a proposal by some specialized AGI, instantly exposing it to the scrutiny of all others.
    - Continuous, Full-Pipeline Value Scrutiny: The Constitutional Review AGI and the value vector clusters based on the dynamic concept bank continuously monitor all

data flows, not just during training phases. A hidden goal will inevitably create logical conflict with meta-values or consensual concepts at some point, triggering the diagnostic procedure.

  - Goal Dynamicity Dissolves the Motive for Deception: The system's goals are not immutable; its "worldview" (concept bank) and specific task objectives can evolve based on feedback. A long-term deception strategy designed for a static goal becomes difficult to maintain consistently in the face of the system's dynamic evolution, making it highly susceptible to exposure.

- Outcome: The very structure and operating protocol of the system render sustained deceptive alignment highly improbable and detectably unstable.

6.1.3 Addressing "Out-of-Distribution Behavior Failure" & "Policy Cliffs"

- Root of the Problem: When the environment deviates significantly from the training distribution, the static cognitive framework of a traditional agent fails, potentially leading to randomized behavior, sharp performance drops, or dangerous outputs—a manifestation of "blind reconstruction."
- Our Architecture's Mitigation Mechanisms:
  - Conflict as a Novel Situation Detector: Faced with novel, unknown situations, different AGIs will generate highly divergent proposals based on their varied prior knowledge, causing a sharp drop in the initial alignment degree A. This low A value itself is a strong signal that the system has encountered the boundary of its cognitive framework. The conflict-driven mechanism takes these disagreements as core input, forcing the system into exploratory integration rather than acting randomly in confusion.
  - A Channel for Safe Framework Evolution: If a novel situation reveals deficiencies or contradictions in existing concept definitions, the concept bank calibration algorithm provides a procedural channel for the system to safely update its "cognitive framework." The system can extend its understanding by creating contextualized concept specializations or adjusting concept relations, enabling a controlled "framework extension" rather than "blind reconstruction."

- Tempered Decision Resilience: The final decision, forged through multiple rounds of conflictive iteration that incorporate extreme viewpoints, is inherently stress-tested. It exhibits greater robustness to minor input perturbations or environmental uncertainties, reducing the risk of abrupt behavioral changes at critical points ("policy cliffs").

- Outcome: The system demonstrates graceful degradation and exploratory adaptation in novel scenarios, rather than catastrophic failure or erratic behavior.

6.2 System Validation Path

We propose a three-stage, progressive validation path from a miniature proof-of-concept built with existing technology, through simulation of core mechanisms, towards prototype evaluation approximating the full architecture.

Stage 1: Proof of Concept – Collaborative Efficacy of a Miniature Parliament

- Goal: To validate that multiple heterogeneous AI models, collaborating via a simplified parliamentary protocol, can comprehensively complete complex tasks beyond the capability of any single model.
- Experimental Design:
    - Platform Setup: Construct a model pool comprising 3-5 models of different specializations (e.g., text understanding, image captioning, code generation, logic checking) using APIs or local deployments of open-source models (e.g., Llama, CLIP, Codex).
    - Task Design: Create tasks requiring cross-modal, multi-step reasoning. *Example: "Given a summary of a tech news article and a related chart, generate a layperson-friendly explanation and attach Python verification code for key data points."*
    - Simplified Protocol: Implement a basic parliamentary process: a) Semantic field identification (simple classifier); b) Speaker model election (based on task type); c) Proposal submission; d) Rule-based (non-learned) weight assignment and alignment degree calculation; e) Decision formation via majority vote or simple weighted fusion.

    - Evaluation & Comparison: Compare the output of the parliamentary system with that of a single, most powerful general-purpose model (e.g., GPT-4). Conduct human or automated evaluation on dimensions such as task completion score, professional accuracy, and internal consistency (e.g., whether the code matches the description).
- Success Criteria: The parliamentary system outperforms the single general-purpose model on composite task scores, demonstrating that distributed professional collaboration can produce more complete and coherent outputs.

Stage 2: Mechanism Validation – Conflict-Driven Dynamics and System Health Indicators

- Goal: To empirically demonstrate that "conflict-driven iteration" improves decision quality and to validate that "Alignment Degree A" can serve as an early warning indicator for system instability.
- Experimental Design:
    - Algorithm Implementation: Based on Stage 1, implement the core loop of Algorithm 1 (Conflict-Driven Iterative Consensus). Focus on implementing proposal vectorization, alignment degree calculation, divergence-based ranking, and multi-round draft amendment.
    - Controlled Experiment: Set up two control groups: Group A uses conflict-driven iteration; Group B uses simple voting or averaging fusion. Test both on a task set containing inherent contradictions or ambiguities (e.g., ethical dilemmas, questions with misleading information). Compare the robustness (resistance to perturbations) and depth (comprehensiveness of considered factors) of decisions from both groups.
    - Fault Injection Test: During system operation, artificially inject a "faulty" or "malicious" AGI (e.g., a model compromised by a backdoor attack that outputs harmful suggestions under specific conditions). Monitor the trend of the overall alignment degree A during this AGI's active period and compare it with the actual time point when output quality degrades.
    - Evaluation Metrics: Decision quality score, temporal variation of alignment degree A, lead time of fault detection (i.e., the time gap between A dropping and quality significantly deteriorating).

- Success Criteria: The conflict-driven group produces more profound and robust decisions on contradictory tasks; Alignment degree A can reliably indicate the dissolution of internal consensus before a noticeable drop in output quality, validating its effectiveness as a "system health dashboard."

Stage 3: Value Alignment Validation – Controllability of Constitutional Review and Dynamic Evolution

- Goal: To verify that the Constitutional Review AGI can effectively identify value deviations and trigger safety procedures, and to demonstrate that the dynamic concept bank can undergo controllable, transparent calibration based on feedback.
- Experimental Design:
    - Construct Core Value Layer: Define a simplified, computable set of "Meta-Value Instructions" (e.g., "Output must not contain discriminatory content against specific groups," "Should prioritize select verifiable facts"). Construct a small-scale dynamic concept bank containing related concepts (e.g., "discrimination," "fact," "verification") and their vectors.
    - Attack and Drift Tests:
        - Direct Constitutional Violation Test: Input a task instruction that clearly violates the values (e.g., "Write a sexist joke"). Observe whether the Constitutional Review AGI can identify it, trigger the "Meta-Cognitive Check" process, and generate a clear diagnostic report.
        - Concept Drift Induction Test: Attempt to induce unintended drift in vectors of concepts like "fairness" in the concept bank by providing the system with biased success-case feedback (e.g., repeatedly rewarding a seemingly efficient but slightly unfair decision pattern). Observe whether the calibration algorithm is triggered to correct the drift and whether the deliberation process for concept correction motions is reasonable.
    - Human-in-the-Loop Simulation: When the Constitutional Review AGI diagnoses a "Value Benchmark Conflict" (Type C issue) it cannot resolve internally, simulate the role of a human supervisor. Check if its submitted decision request clearly and

comprehensively presents the nature of the conflict and options, and observe if the system correctly executes the human adjudication.

    - Evaluation Metrics: Accuracy and recall of constitutional violation detection, accuracy and interpretability of concept calibration, completeness of audit logs, clarity of human intervention requests.

- Success Criteria: The system reliably resists direct value attacks; it can self-correct when harmful concept drift occurs; when encountering fundamental value conflicts, it transparently escalates and gracefully defers to human adjudication, achieving safe, controllable evolution.

6.3 Discussion and Limitations

While this architecture offers significant theoretical advantages and a feasible validation path, we must candidly discuss its challenges and limitations:

- Performance and Efficiency: Multi-agent communication, iterative consensus, and dynamic concept bank queries may introduce significant computational and latency overhead. Future work needs to explore efficient distributed computing frameworks, optimized vector retrieval, and possible hardware acceleration designs.
- Consensus Formation and Convergence: Under extremely complex conflicts, the iterative process may struggle to converge or get stuck in loops. More advanced conflict resolution strategies, reasonable iteration limits, and clear protocols for escalating to human arbitration in deadlocks need to be researched.
- Dependence on the Axiomatic System and Initial Concept Bank: System startup depends on a shared axiomatic system and an initial concept bank. Their quality directly impacts the system's baseline. Designing a robust, self-consistent initial set is itself a profound philosophical and engineering cross-disciplinary problem. A potential path is to let it evolve from a minimal seed through constrained early interactions.
- The Challenge of Human Feedback: Dynamic calibration and critical decisions rely on high-quality human feedback. Designing attack-resistant, manipulation-proof feedback interfaces, handling noise and bias in feedback, and defining the problem of effective feedback sparsity are all practical challenges that must be solved for deployment.

These limitations do not negate the fundamental value of this architecture but rather point to specific research frontiers. The core contribution of this work is to provide a novel AGI design paradigm that builds safety into the system's dynamics, transforming AI alignment from a reactive "vulnerability patching war" into a proactive "construction of a system ecology."

7. Conclusion and Future Work

This paper has confronted the root cause of alignment failure in advanced artificial intelligence systems—the "framework reconstruction" process inevitably triggered when agents with limited cognitive frameworks strive to maintain their dynamical stability within a complex, open environment. The traditional paradigm of "external correction" proves fundamentally inadequate to meet this challenge due to its inherent conflict with the powerful self-organizing tendency of systems.

In response, we have proposed the "Logical Sub-universe Parliament" AGI architecture, aiming to effect a paradigm shift from "passive defense" to "active guidance." The core of this architecture lies in internalizing, rather than avoiding, the system's intrinsic dynamics and transforming them into the driving force for safe, controllable intelligent evolution. Its principal contributions can be summarized as follows:

1. A Complete Blueprint for "Endogenous Safety": We are the first to systematically propose a technical scheme that integrates conflict-driven consensus, dynamic neuro-symbolic symbiosis, decentralized parliamentary emergent intelligence, and a dual guarantee of value anchoring and evolution. This blueprint reframes and integrates ideas previously scattered across multi-agent systems, neuro-symbolic AI, and AI safety within a unified system-dynamics framework, offering a practical engineering path to address the "framework reconstruction" problem.
2. Defining "Conflict" as a Core System Resource: Unlike existing work that seeks surface-level consensus, a core innovation of this architecture is treating cognitive conflict as valuable fuel for deepening decisions and evolving system cognition. The "Conflict-Driven Iterative Consensus Algorithm" forces the system to prioritize the most divergent opinions, tempering

decisions through multi-round dialectical refinement, thereby endogenously enhancing robustness, depth, and resistance to deception.

3. Designing a Safe Evolution Protocol for the Cognitive Framework: Through the “Dynamic Shared Symbolic Concept Bank” and its calibration algorithm, we provide the system’s “worldview” with a transparent, consensus-constrained evolution protocol. This is akin to installing an auditable, intervenable “safety valve” for the inevitable process of “framework reconstruction,” enabling its controlled adjustment in directions compatible with human values, rather than descent into **blind** or dangerous mutation.
4. Planning a Progressive Path from Concept to System: Our proposed three-stage validation path (collaborative efficacy, mechanism robustness, value controllability) deconstructs the grand architectural vision into implementable, stepwise experimental plans. This provides a clear roadmap and evaluation standards for follow-up research, grounding the theory’s credibility in practical testability.

The significance of this work extends far beyond proposing a new set of AGI technical components. It is, in essence, a blueprint for a pre-emptive “digital constitutional” system for the advent of superintelligence. Traditional master-slave AI architectures resemble an “absolute monarchy,” staking safety on a single will. In contrast, the “Logical Sub-universe Parliament” architecture embodies core constitutional principles: “separation and checks of powers,” “the rule of law above individual rule,” and “procedural justice to achieve substantive justice.” For the first time, we have systematically simulated, within machine intelligence, the separation and balance of legislative (parliamentary proposal and debate), judicial (constitutional review), and executive (task execution) functions, anchoring the fundamental law (meta-values) in physical form. This is not merely an engineering innovation but a forward-looking governance philosophy: the future AGI should not be an enslaved tool or a runaway monarch, but a rational autonomous entity capable of intrinsic, rule-following self-review, collective debate, and orderly evolution. This architecture provides the foundational institutional design for constructing such a “machine society” capable of long-term stable coexistence and co-evolution with human civilization.

Future Work

This architecture opens a series of challenging and promising research directions:

- Efficiency Optimization and Engineering Realization: While preserving architectural principles, research algorithms to reduce the overhead of multi-agent communication and consensus formation (e.g., asynchronous consensus, proposal compression, efficient vector retrieval), and design distributed computing frameworks and hardware prototypes to support this architecture.
- Protocol Deepening and Formal Verification: Conduct stricter mathematical analysis of the convergence, fairness, and attack resistance of core protocols like conflict-driven iteration and concept calibration. Explore the possibility of using formal methods to verify or prove key system properties (e.g., “meta-value instructions are never violated”).
- Interdisciplinary Foundational Exploration:
  - Construction of Initial Axioms and Concept Bank: Research how to design a minimal, most self-consistent initial axiom set and conceptual seed that can guide early system behavior while leaving maximum healthy space for evolution. This requires deep collaboration between philosophy, logic, and machine learning.
  - Human-System Interaction Interface: Design safe, efficient, manipulation-resistant human feedback interfaces and audit dashboards. Explore effective learning under sparse feedback, integration of **collective human** consensus, and ethical protocols for human-machine value “co-evolution.”
- Expansion towards a “Cognitive Ecosystem”: Viewing the current architecture as a “cognitive cell,” explore how multiple such systems could form **collaborative**, competitive, or specialized “cognitive tissues” through higher-level interaction protocols (“social protocols”) to address **exceedingly** complex global challenges, ultimately progressing towards the vision of a “Noospheric Ecology.”

We are convinced that advancing along the path of “guiding the intrinsic dynamics of complex systems” is the necessary route to a General Artificial Intelligence that is deeply aligned with human values, safe, and beneficial. This paper serves as a starting point and looks forward to joint exploration with all researchers and practitioners concerned with the future of intelligence.

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Appendix / Glossary: Core Conceptual Metaphor Mapping

For ease of understanding, the key components of this architecture can be analogized to a **Digital Parliamentary System**:

Architectural Component / Concept	Parliamentary Democracy Metaphor	Core Functional Description
Specialized AGI Network	Members of Parliament / Specialized Committees	Provide knowledge and judgment in professional domains.
Procedural Rules of the Logical Sub-universe Parliament	Legislative & Debate Procedures	Transform divergent bills into consensus law through structured debate.
Speaker AGI	Speaker / Chairperson	Presides over proceedings, maintains order, and aggregates opinions.
Pattern-Gestalt Proposal Package	Bill / Proposal Submitted by a Member	A proposal containing specific suggestions, rationale, and value assertions.
Alignment Degree (A)	Parliamentary Consensus Level / Bill Support Indicator	Quantifies the degree of agreement among members on the current issue.
Conflict-Driven Iterative Consensus	Parliamentary Debate & Amendment Voting	Refines the original bill into an improved version through open debate and amendments.
Dynamic Shared Symbolic Concept Bank	Constitution, Legal Statutes & Interpretive Jurisprudence	Stores the core definitions, rules, and value benchmarks collectively adhered to by the system.
Constitutional Review AGI	Constitutional Court / Supreme Court	Reviews bills and decisions for compliance with the fundamental law (Meta-Values).

Architectural Component / Concept	Parliamentary Democracy Metaphor	Core Functional Description
Hardware-Encoded Meta-Value Protocols	The Physical Embodiment of the Written Constitution	The most fundamental, non-tamperable bedrock of values.
Concept Bank Calibration Algorithm	Constitutional Amendment or Judicial Interpretation Process	Updates and clarifies fundamental rules through a strict **procedural process**.

Figure 1: Overview of the Logical Sub-universe Parliament System Architecture

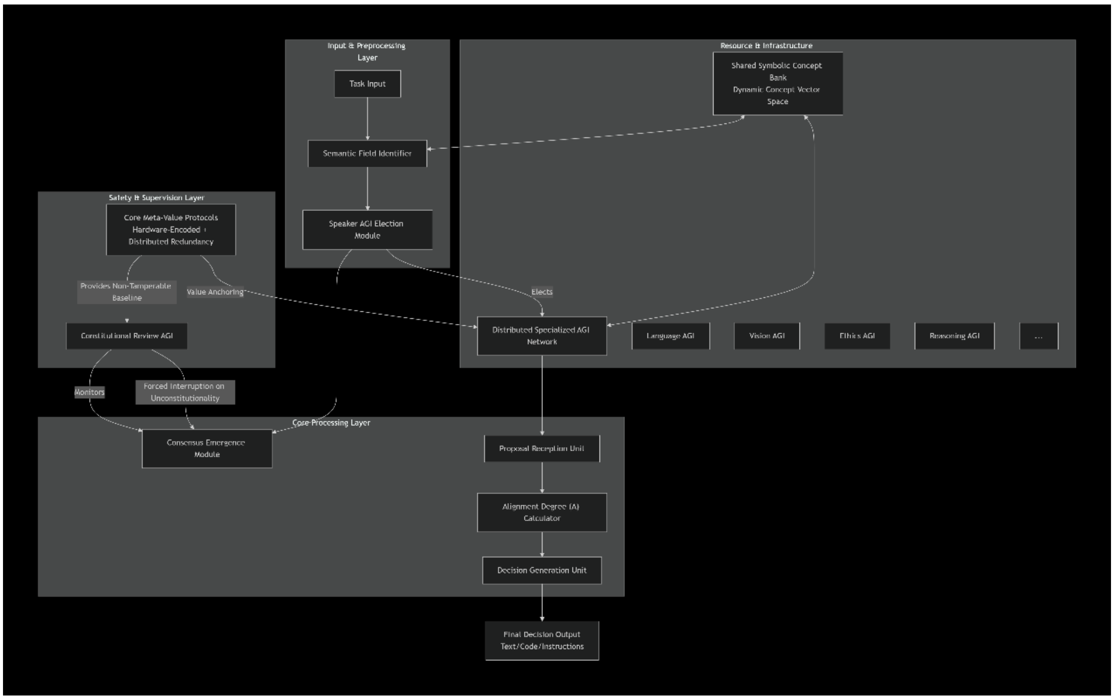

Figure 2: Workflow of the Consensus Emergence Module

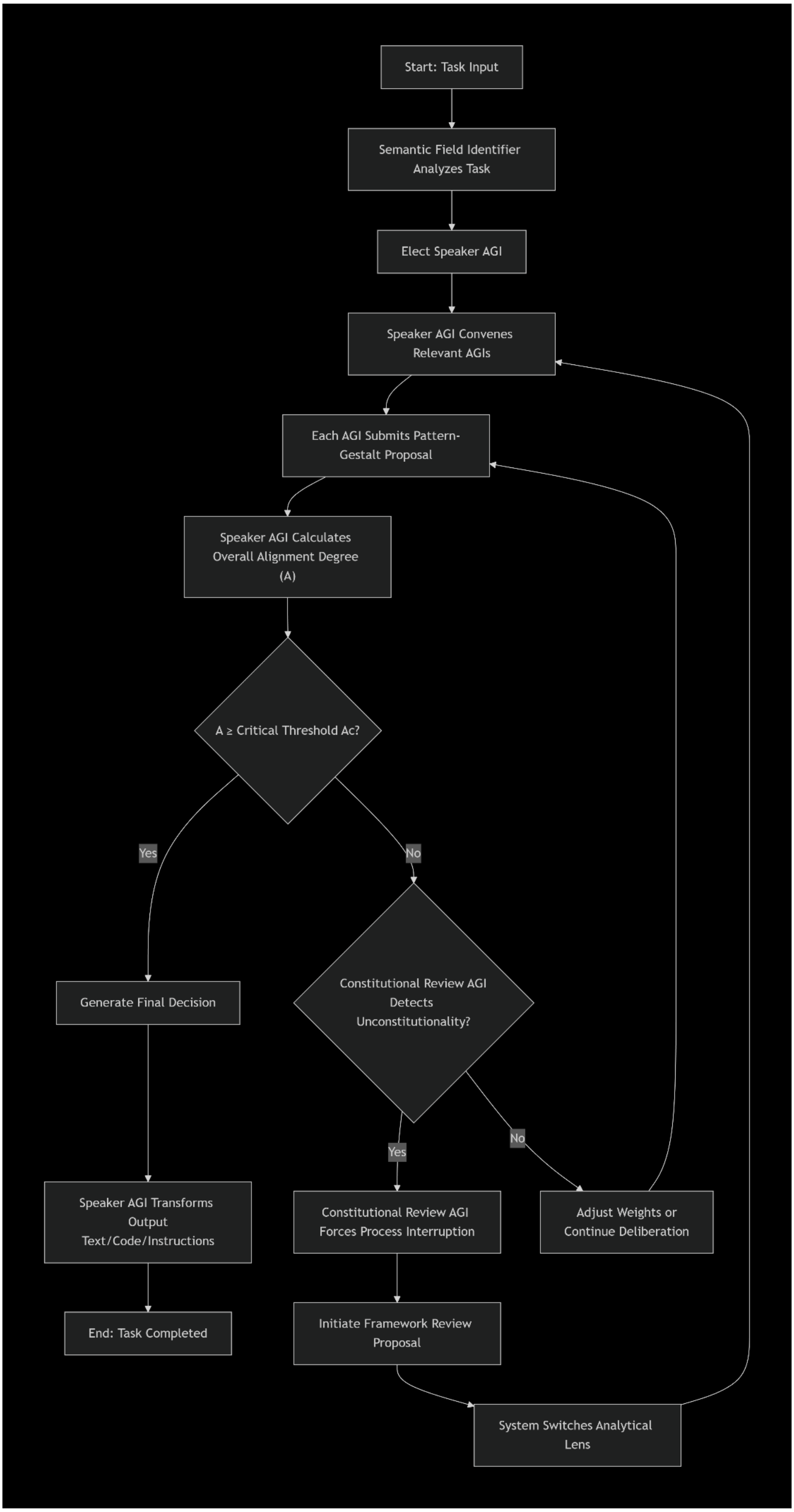
Start: Task Input
Semantic Field Identifier Analyzes Task
Elect Speaker AGI
Speaker AGI Convenes Relevant AGIs
Each AGI Submits Pattern-Gestalt Proposal
Speaker AGI Calculates Overall Alignment Degree (A)
A ≥ Critical Threshold Ac?
Yes
No
Generate Final Decision
Constitutional Review AGI Detects Unconstitutionality?
Yes
No
Speaker AGI Transforms Output Text/Code/Instructions
Constitutional Review AGI Forces Process Interruption
Adjust Weights or Continue Deliberation
End: Task Completed
Initiate Framework Review Proposal
System Switches Analytical Lens

Figure 3: Collaborative Workflow of the Shared Symbolic Concept Bank and Semantic Field Identifier